THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 2: **Electricity: Circuits and their Components** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Overloading of a circuit can cause: 1

(a) fire **(b)** wind **(c)** rain **(d)** cold

2. A bulb glows when: 1

(a) the cell is removed **(b)** current passes through its filament

(c) the switch is off **(d)** the wire is cut

3. The symbol of a bulb in a circuit diagram is: 1

(a) two parallel lines **(b)** a zigzag

(c) a long line **(d)** a circle with a cross inside

4. Short circuit happens when: 1

(a) the switch is off **(b)** the bulb is fused

(c) live and neutral wires touch **(d)** the cell is dead

5. In a cell, the longer metal cap is the: 1

(a) positive terminal **(b)** switch

(c) fuse **(d)** negative terminal

6. The symbol for an open switch is a: 1

(a) gap with a slanted line **(b)** closed straight line

(c) zigzag **(d)** circle

7. An LED glows when current passes through it in: 1

(a) no direction **(b)** only one direction

(c) either direction **(d)** all directions

8. The flat metal base of a cell is the: 1

(a) filament **(b)** negative terminal

(c) switch **(d)** positive terminal

9. A diagram that uses symbols to show an electric circuit is a: 1

(a) map **(b)** circuit diagram **(c)** wiring **(d)** chart

10. Compared to an ordinary bulb, an LED: 1

(a) gets very hot **(b)** uses more energy

(c) uses much less energy **(d)** has a filament

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SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why should you not touch electrical appliances with wet hands? 2

12. What is a fuse and what does it do? 2

13. Why are electric wires made of copper? 2

14. What is an MCB? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Why are the handles of electrician's tools covered with rubber or plastic? 3

16. Explain what happens if a fuse of too high rating is used. 3

17. Name three safety devices used in homes. 3

OR

Describe the activity to test conductors and insulators using a circuit. 3

18. Explain how an LED differs from a bulb. 3

OR

Why is the human body a conductor? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the structure of a fuse and an MCB, and explain why they are used. 5

OR

Explain why an electric iron becomes hot but the cord does not. 5

20. Explain the use of symbols and draw (describe) a circuit diagram with two cells, a bulb and a switch. 5

OR

Describe the role of LED lights in saving energy. 5

END OF QUESTION PAPER — Total: 40 Marks

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